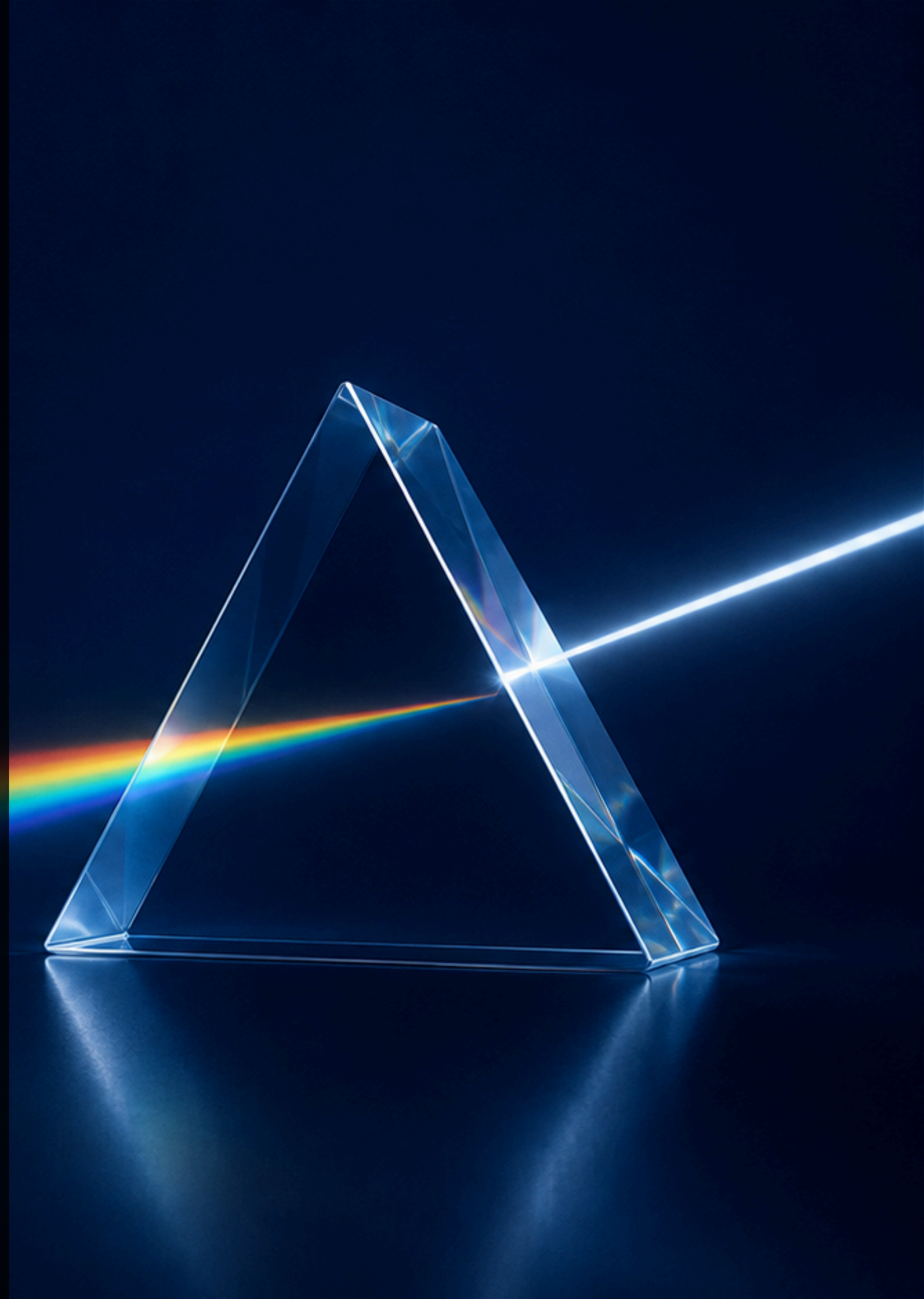


Black-body radiation

Emission spectrum · Stefan–Boltzmann law · Wien's law

Laboratory of Modern Physics
Giuseppe Frisella · Scienze Fisiche
Università degli Studi di Palermo · 2025/2026



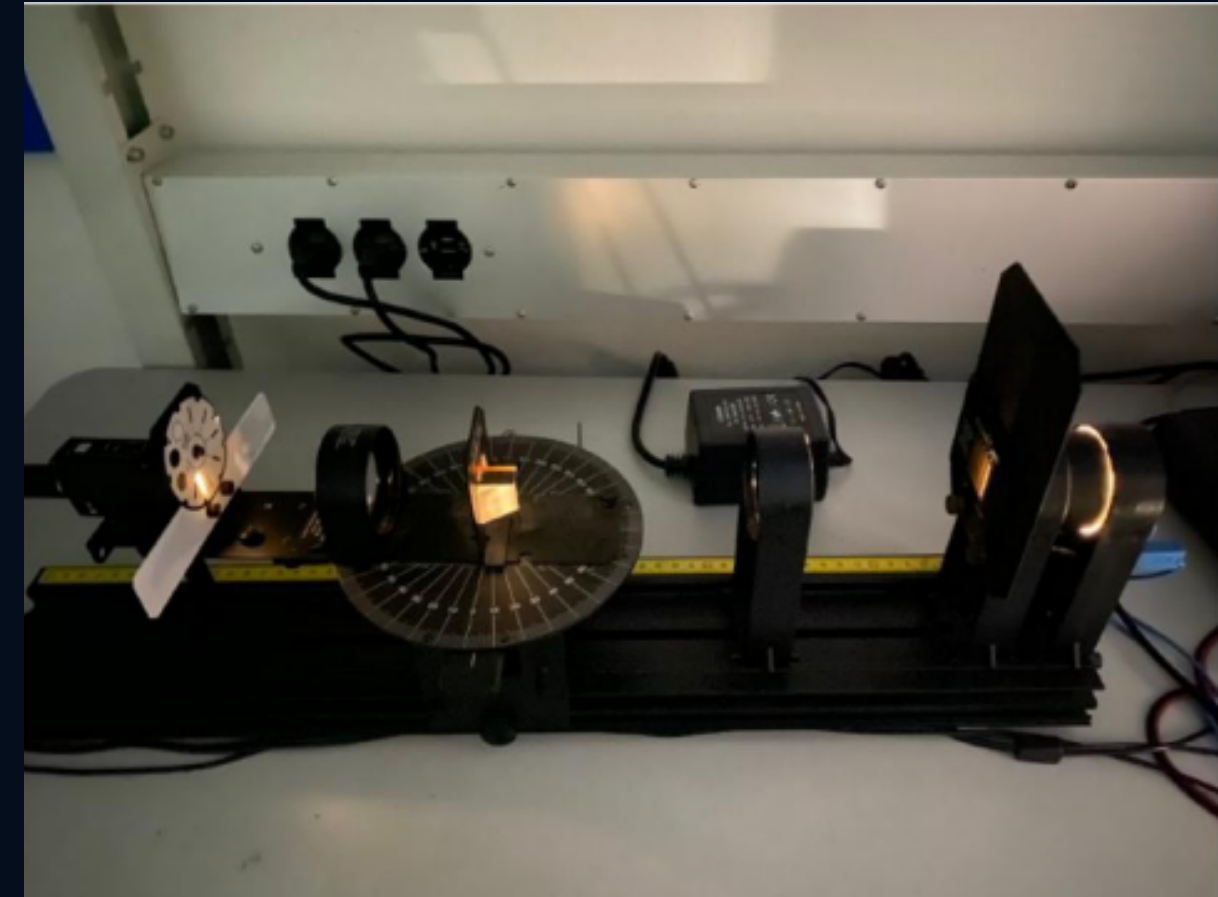
1. Target and apparatus

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} \left[\exp \left(\frac{hc}{\lambda k_B T} \right) - 1 \right]^{-1}$$

$$A = \int I_{\text{rel}}(\lambda) d\lambda \propto T^4 \quad \lambda_{\text{max}} T = b_{\text{Wien}}$$

Experimental chain

Heated filament → prism spectrometer → rotating detector



Processing Pipeline

Angle → wavelength conversion → baseline subtraction →
integration and local estimate

2. From raw angle to wavelength

$$\theta = \theta_0 - \frac{\theta_{\text{raw}}}{R} \quad n(\theta) = \sqrt{(1.1547 \sin \theta + 0.5)^2 + 0.75} \quad \lambda = \sqrt{\frac{13900}{n(\theta) - 1.689}}$$

Only the 300–1000 nm interval was kept, where calibration and response are more reliable

Numerical Analysis

$$A \simeq \sum_i \frac{I_i + I_{i+1}}{2} (\lambda_{i+1} - \lambda_i)$$

$$A = aT^4 + b$$

$$\lambda_{\text{max}} = \frac{b_{\text{fit}}}{T} + q$$

Key limitation

A is relative, not watts: no absolute radiometric calibration was available

3. Temperature and uncertainty

Temperature from resistance

$$R_f = \frac{V}{I} \quad T = T_0 + \frac{R_f/R_0 - 1}{\alpha}$$

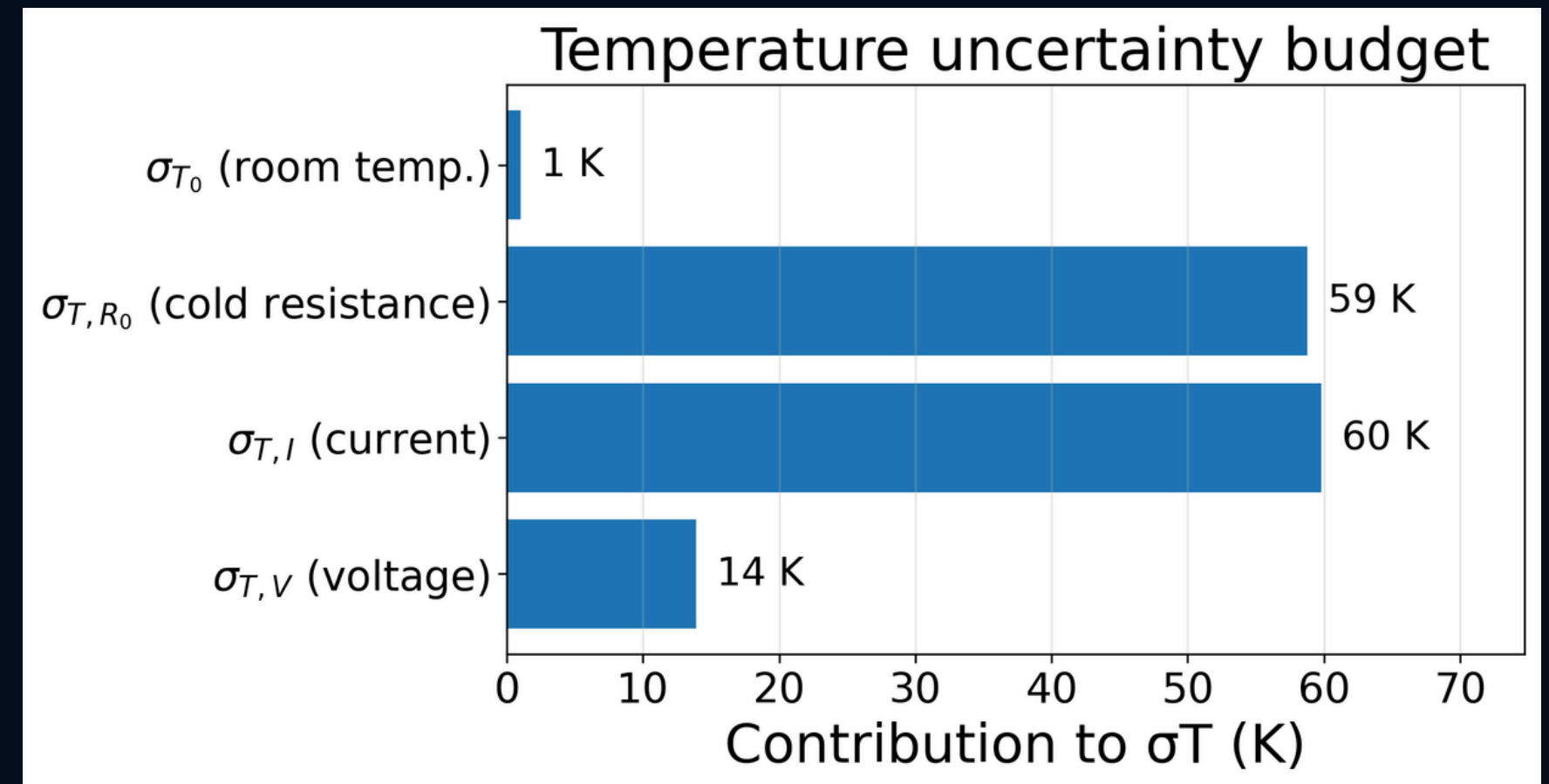
$$\sigma_T^2 = \left(\frac{\partial T}{\partial V} \sigma_V\right)^2 + \left(\frac{\partial T}{\partial I} \sigma_I\right)^2 + \left(\frac{\partial T}{\partial R_0} \sigma_{R_0}\right)^2 + \sigma_{T_0}^2$$

Error origin

$$\sigma_V = 0.3 \text{ V} \cdot \sigma_I = 0.02 \text{ A}$$

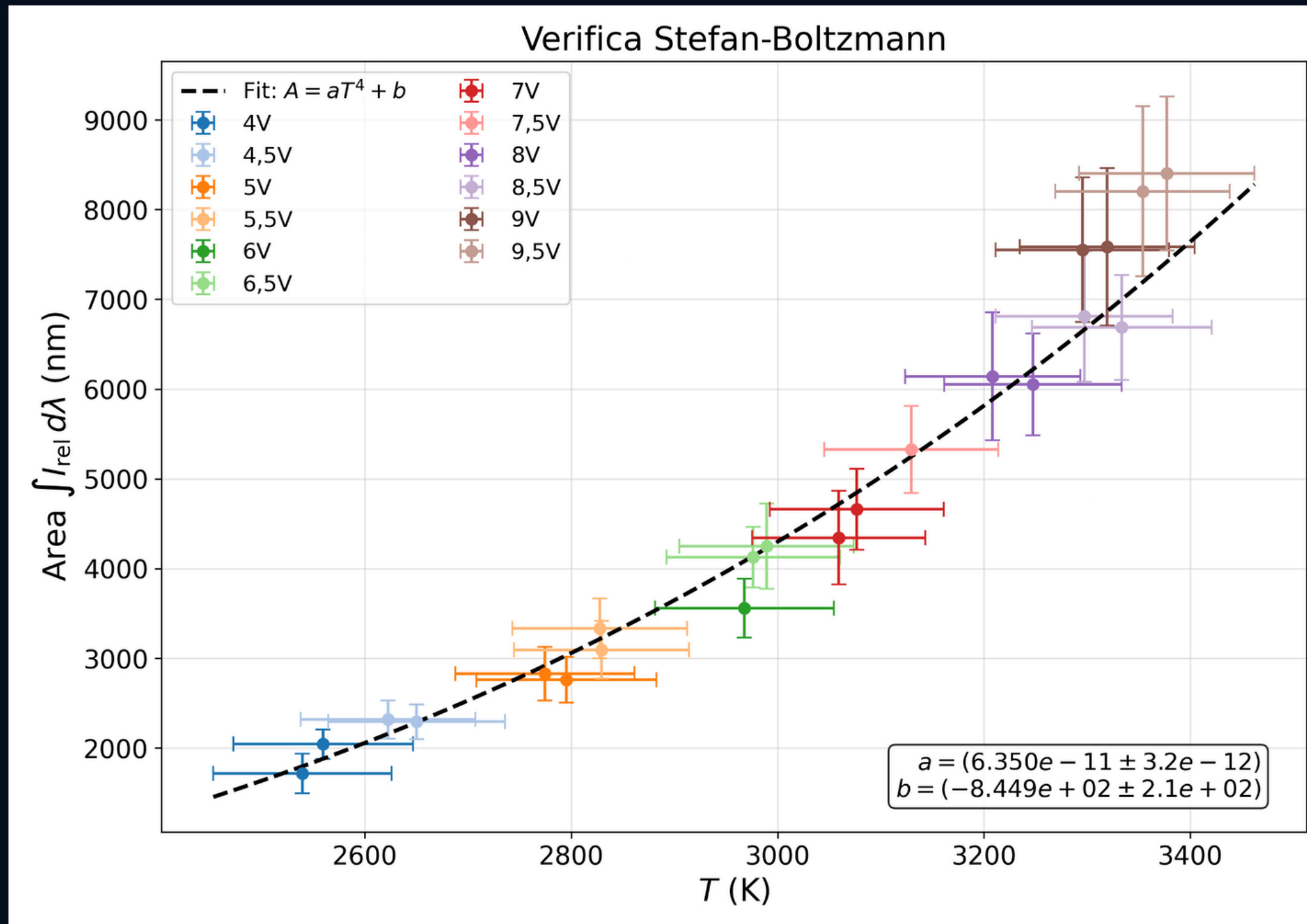
$$\sigma_{R_0} = 0.1 \text{ } \Omega \cdot \sigma_{T_0} = 1 \text{ K}$$

Mean full $\sigma T \approx 85 \text{ K}$



Current and cold-resistance dominate σT

4. Stefan–Boltzmann law



$$A = aT^4 + b$$

Fit parameters

$$a = (6.350 \pm 0.321) \times 10^{-11}$$

$$b = -845 \pm 208$$

Goodness

$$\chi^2/\text{dof} \approx 0.57$$

The integrated spectrum follows a T^4 -like trend

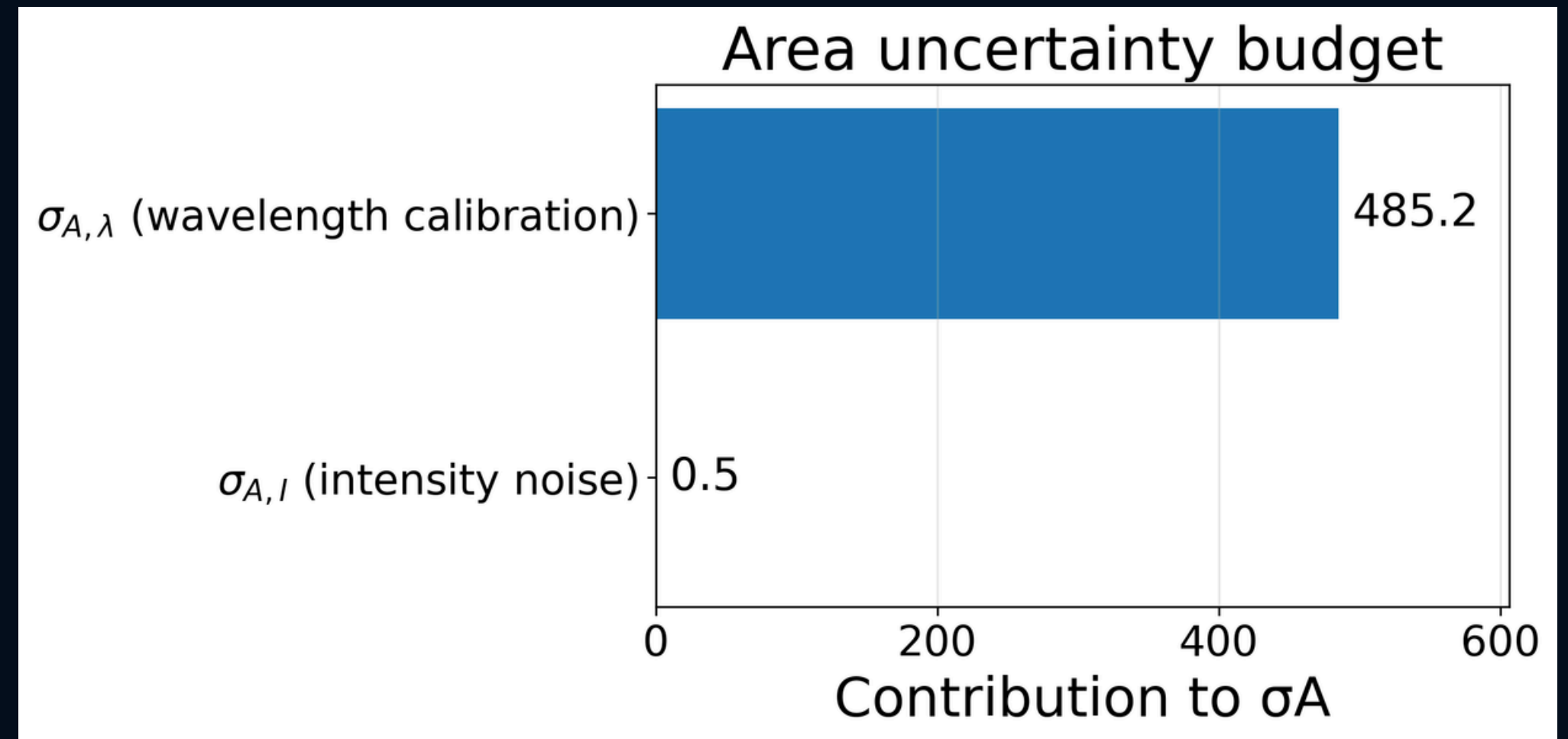
5. Area uncertainty

Error bars

Intensity: noise of the spectrum.

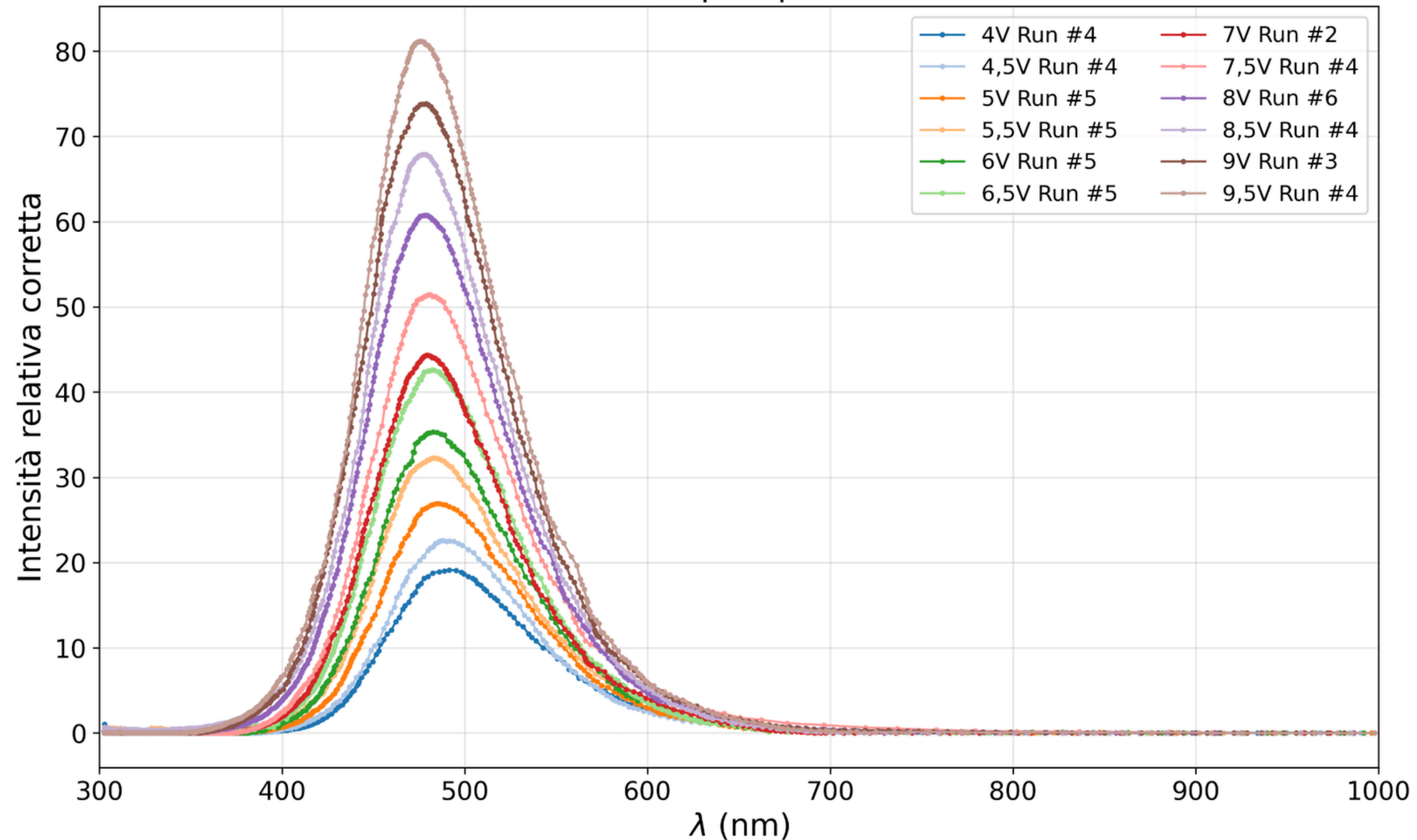
Wavelength: propagated angle $\rightarrow \lambda$ uncertainty through each trapezoid

$$\sigma_A = \sqrt{\sigma_{A,I}^2 + \sigma_{A,\lambda}^2}$$



6. Measured spectra

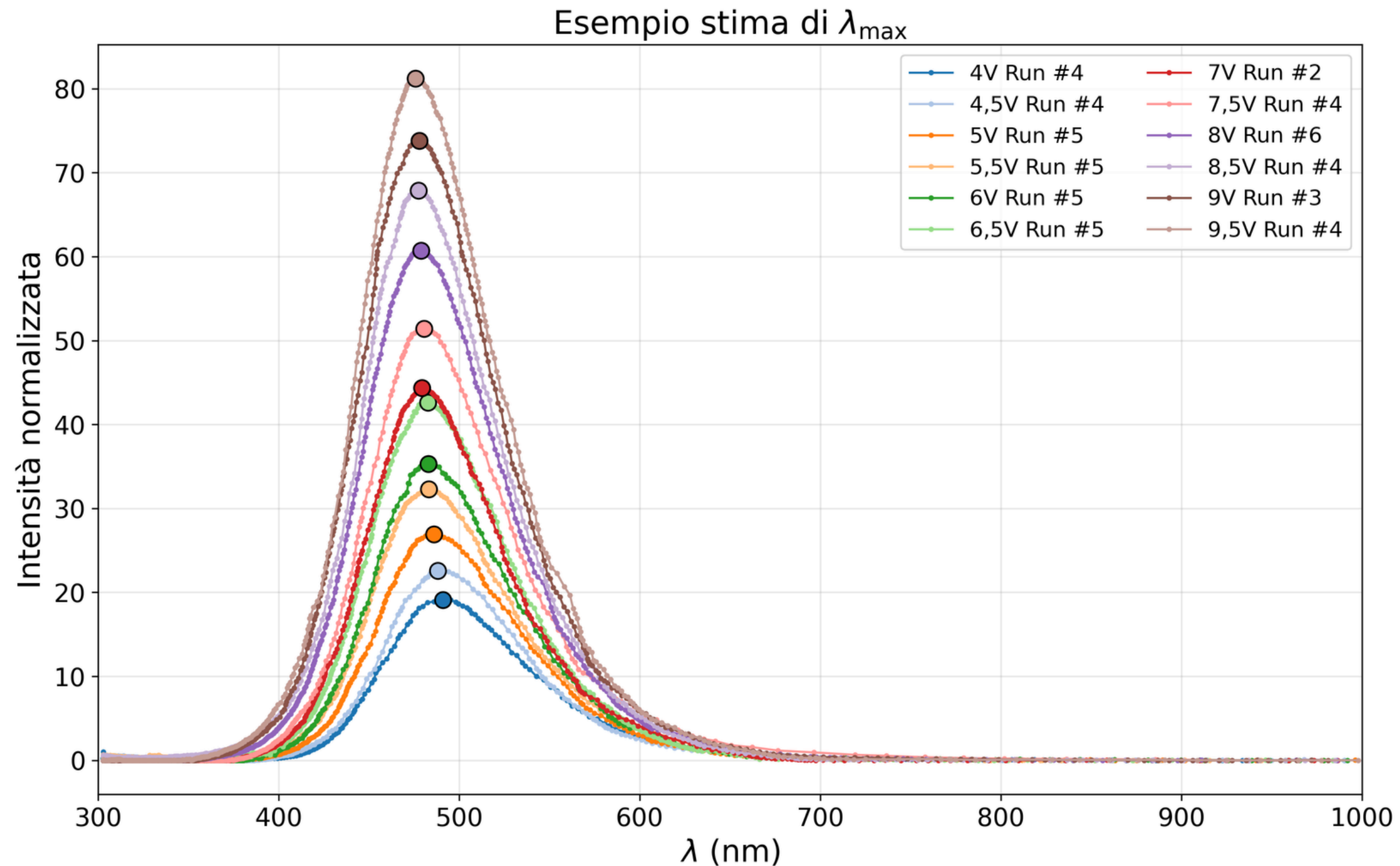
Esempio spettri



Reading the plot

Area: the integrated signal grows by about 4.9 times

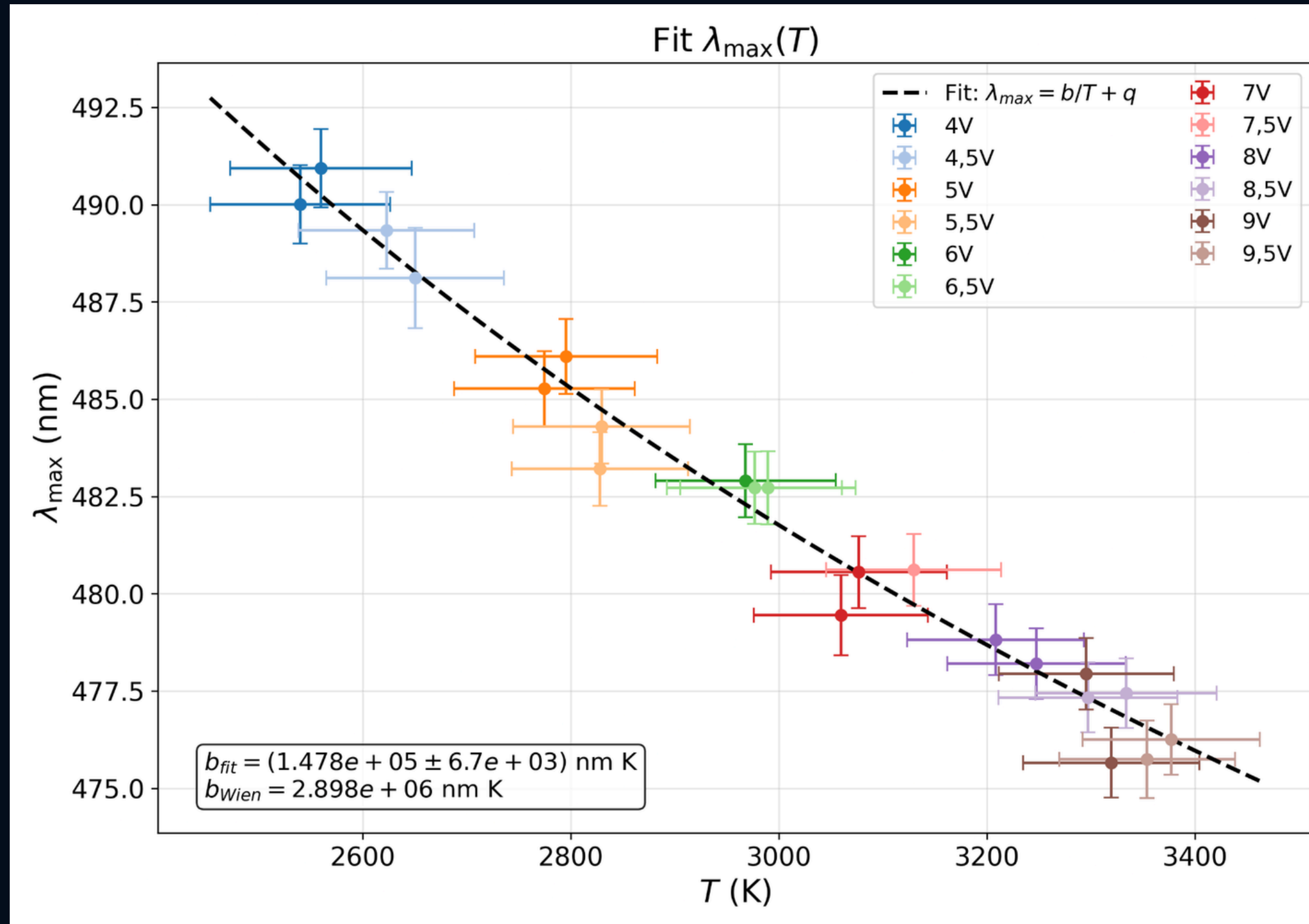
7. Local $\lambda_{\max}$ estimate



Reading the plot

Locate the peak → Fit a parabola → find the vertex to evaluate $\lambda_{\max}$

8. Wien displacement law



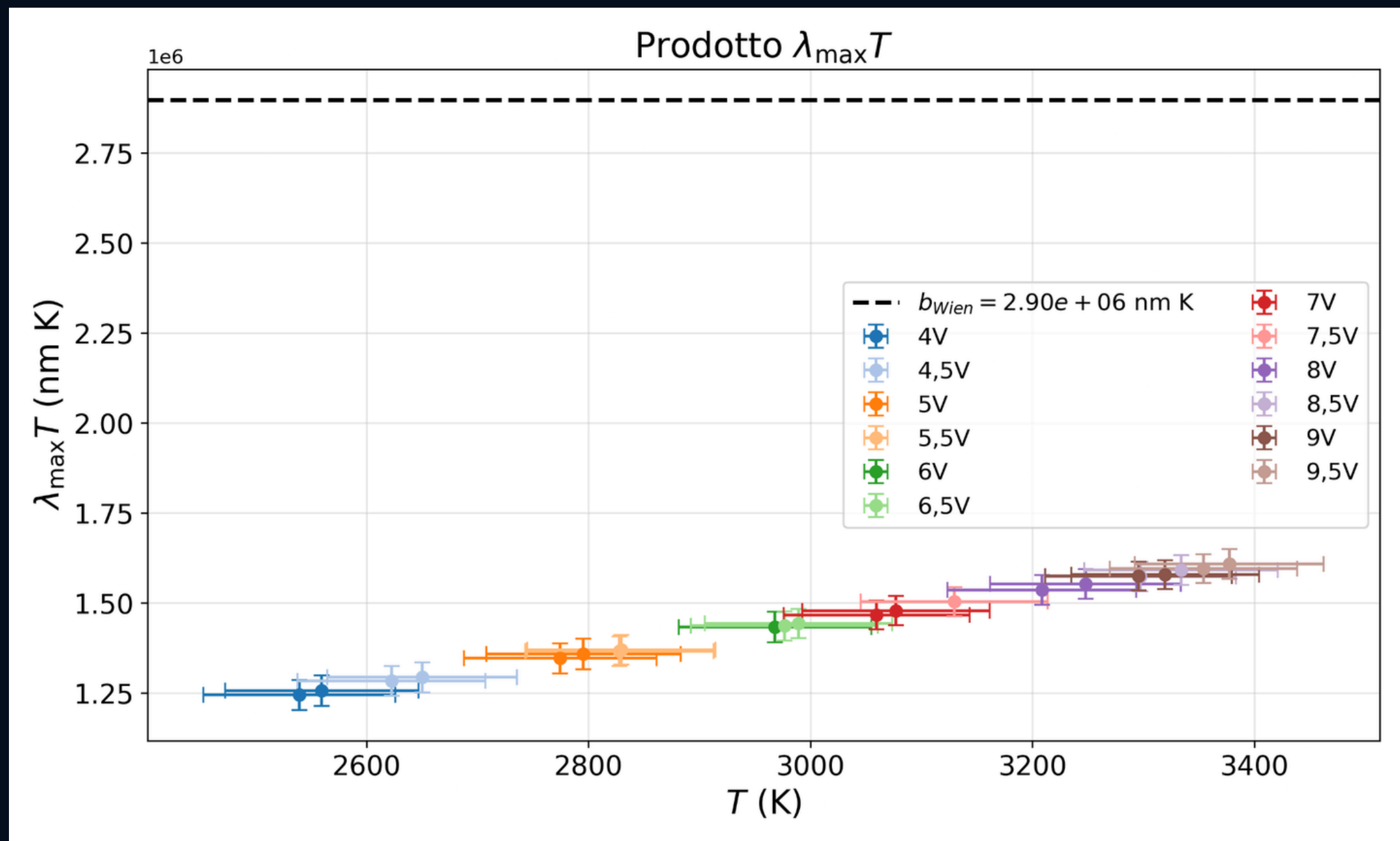
$$\lambda_{\max} = \frac{b_{\text{fit}}}{T} + q$$

Fit result

$b_{\text{fit}} = (1.478 \pm 0.067) \times 10^5 \text{ nm K}$
 $q = 432.51 \pm 2.24 \text{ nm}$

The $1/T$ trend is visible, but the fitted coefficient is not the ideal Wien constant

9. Critical check: $\lambda_{\max}T$

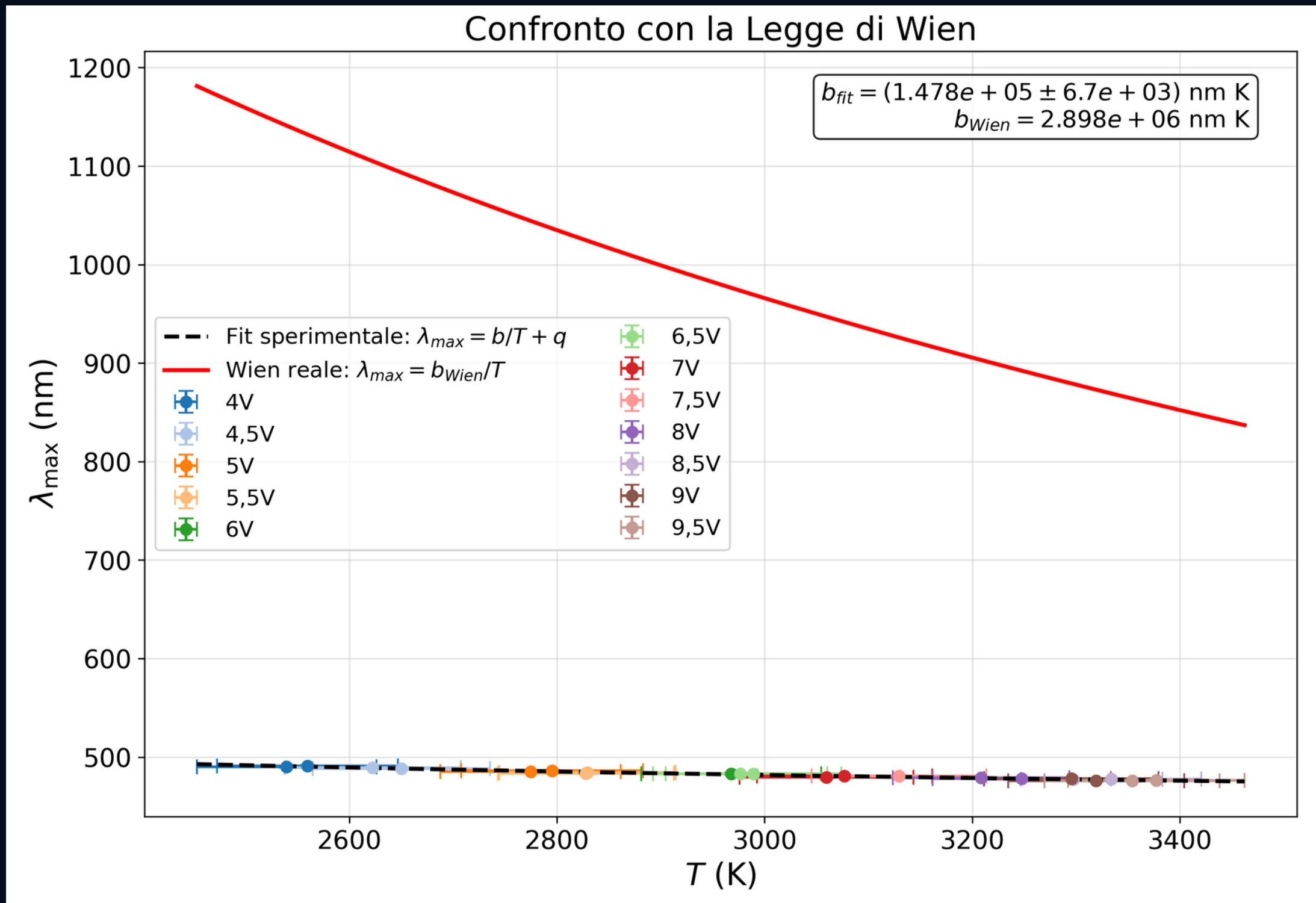


Values

Data: $\lambda_{\max}T \approx 1.25\text{--}1.60 \times 10^6$ nm K
Theory: $b_{Wien} = 2.8978 \times 10^6$ nm K
Ratio: $b_{fit}/b_{Wien} = 0.051$

The measured maximum is apparatus-shaped: lamp non-ideality, prism response and detector sensitivity matter

10. Critical check: $\lambda_{\max}$ as a function of T



Results

The measured peaks stay near 475–491 nm, while it should be much higher in this temperature range

11. Final conclusions

Stefan-Boltzmann

The integrated area grows with T^4

Wien

The peak shifts in the right way, but the ideal constant is not recovered

Main takeaway: absolute constants require absolute spectra

